

(ii) 0%

A1 [2]

2 (a) all same speed in a vacuum (allow medium)/all travel in a vacuum (1)

transverse/can be polarised (1)

undergo diffraction/interference/superposition (1)

can be reflected/refracted (1)

show properties of particles (1)

oscillating electric and magnetic fields (1)

transfer energy/progressive (1)

not affected by electric and magnetic fields (1)

(allow any three, 1 each)

B3 [3]

(b) $495 \text{ nm} = 495 \times 10^{-9} \text{ m}$

C1

number = $1/(495 \times 10^{-9}) = 2.02 \times 10^6$

A1 [2]

(allow 2 or more significant figures)

(c) (i) allow $10^{-7} \rightarrow 10^{-11} \text{ m}$

B1

(ii) allow $10^{-3} \rightarrow 10^{-6} \text{ m}$

B1 [2]